

Understanding the Technology

Estimation method for temperature uncertainty of temperature chambers (JTM K 08)

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In July 2007, The Japan Testing Machinery Association (JTM) issued a new standard for evaluating temperature uncertainty for temperatures implemented in temperature test chambers, JTM K 08 "Estimation method for temperature uncertainty of temperature chambers" (below K 08). This standard considers uncertainty based on JTM K 07 "Temperature chambers – Test and indication method for performance" (below K 07) issued in March 2007. This article will provide a commentary on understanding and calculating temperature uncertainty according to JTM K 08, as well as mentioning points of debate and points in which unification of standards has not been possible.

1 Introduction

The ISO/IEC standard 17025 "General requirements for the competence of testing and calibration laboratories" requires testing laboratories and calibration organizations to maintain a procedure for estimating measurement performance. Evaluating the level of temperature performance achieved within the test chambers for temperature testing, commonly referred to as environmental testing, has become mandatory.

Because of this, in addition to establishing K 07, JTM has enacted K 08, "Estimation method (and procedure) for temperature uncertainty of temperature chambers."

Espec Technology Report No.24 (September 1, 2007), provided a detailed analysis of K 07, which entails major differences with previous JTM standards (K 01, 03, 05). Temperature performance measured within the test chambers consists of the maximum value of the temperature fluctuation, temperature gradient, and temperature variation in space, which is expressed as test chamber performance. In other words, this value indicates the size of the temperature performance dispersion. Therefore, the uncertainty of the temperature dispersion can be calculated based on K 07.

2 Summary of test chamber temperature performance

2-1 Definition of temperature performance, and method of expressing

This overlaps somewhat with Espec Technology Report No.24, but Table 1 presents the main points concerning temperature performance as stipulated in K 07.

Table 1 Points of difference between JTM K 07 and previous JTM standards

Previous standards (JTM K 05)		New standards (JTM K 07)	
Performance item	Definition and method of determining	Performance item	Definition and method of determining
Working space	Space inside chamber excluding 1/6 the distance from walls	Working space	Space inside chamber excluding 1/10 the distance from walls
Range of temperature fluctuation	Measurement points: 1 point (center of chamber only) Calculation method: Find average value of data obtained. Separate the same data into values below average and above average, then find the average value of the larger group (average maximum temperature) and the smaller group (average minimum temperature). Divide the difference between these two values by 2 and display as $\pm$ value.	Temperature fluctuation	Measurement points: 9 points (center of chamber and 8 corner points of working space) Calculation method: Find standard deviation σ from measurement data taken at each measurement point. Use greatest σ value of the 9 obtained, multiply by 2, and display as $\pm$ value.
Temperature uniformity	Measurement points: 9 points (center of chamber and 8 corner points of working space) Calculation method: Find the average temperature of chamber center. Separate temperature measurements taken at 8 corners into greater and smaller than chamber center average. Find the average values for each group, divide the difference between these averages by 2, and display as $\pm$ temperature uniformity value.	Temperature gradient	Measurement points: 9 points (center of chamber and 8 corner points of working space) Calculation method: Find the average temperature at each of the 9 measurement points. Find the difference between the greatest and smallest values from these 9 points and display as the temperature gradient.
		Temperature variation in space	Measurement points: 9 points (center of chamber and 8 corner points of working space) Calculation method: Find the average temperature at each of the 9 measurement points. Find the absolute value of the difference between the average temperature of chamber center and average temperatures of the 8 corners and display the maximum value as the temperature variation in space.

2-2 Working space

Fig.1 shows the working space. The outer solid line dimensions represent the inside surface of the chamber walls, while the area enclosed by the inner gray surfaces represents the working space.

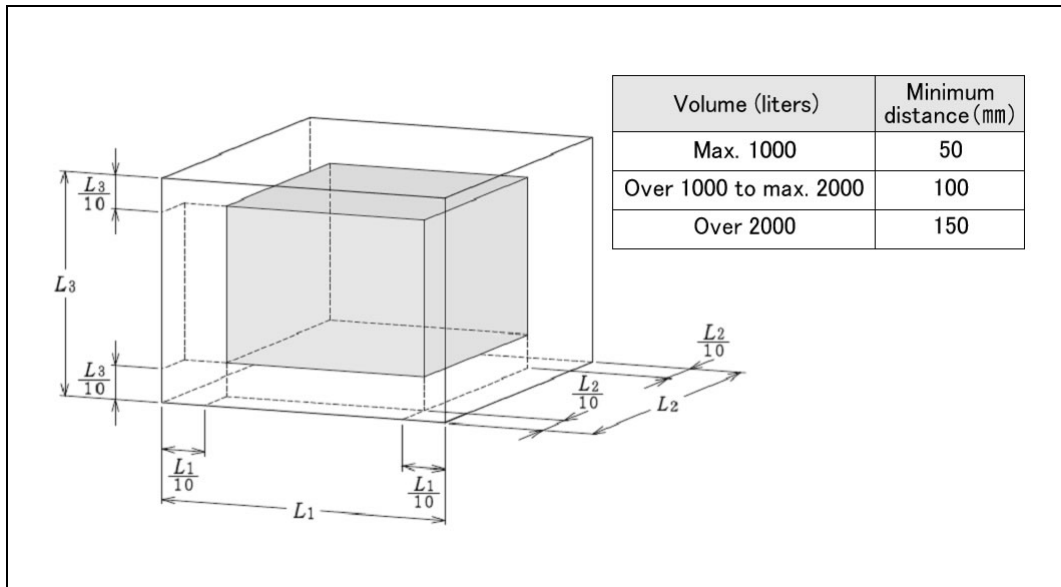


Fig.1 Working space

3 Test chamber temperature and uncertainty

3-1 Temperature performance and uncertainty

As can be seen in Table 1, with K 07 the temperature fluctuation, temperature gradient, and temperature variation in space are obtained by measuring a total of nine points (in the chamber center and the eight corners) and expressing the temperature performance according to the greatest values. In addition, the size of the working space considered for determining performance has increased, with temperature uniformity being considered closer to the chamber walls than in the previous standards, which emphasized average performance within a narrower working space.

K 07 expresses the temperature dispersion relative to time and space, and if the temperature deviates at even one point among the nine measurement points or fluctuates in time, that will affect the performance evaluation. This is a very strict provision, but it provides an estimate of uncertainty based on actual temperature dispersion.

3-2 Test chamber representative temperature

When considering test chamber temperature, the most basic fact to recognize is that the test chamber temperature depends on which area of the test chamber is specified. The JTM standards previously specified the test chamber temperature as the temperature at the geometrical center of the chamber (chamber center). This new standard (K 08) can also be seen as considering the temperature at the center of the chamber to be the representative temperature.

However, as Fig.2 shows, the temperature controllers installed in the test chamber to measure and control temperature are often located near the intake and blowout ports of the air circulators that circulate the air inside the test chambers and the chamber corners.

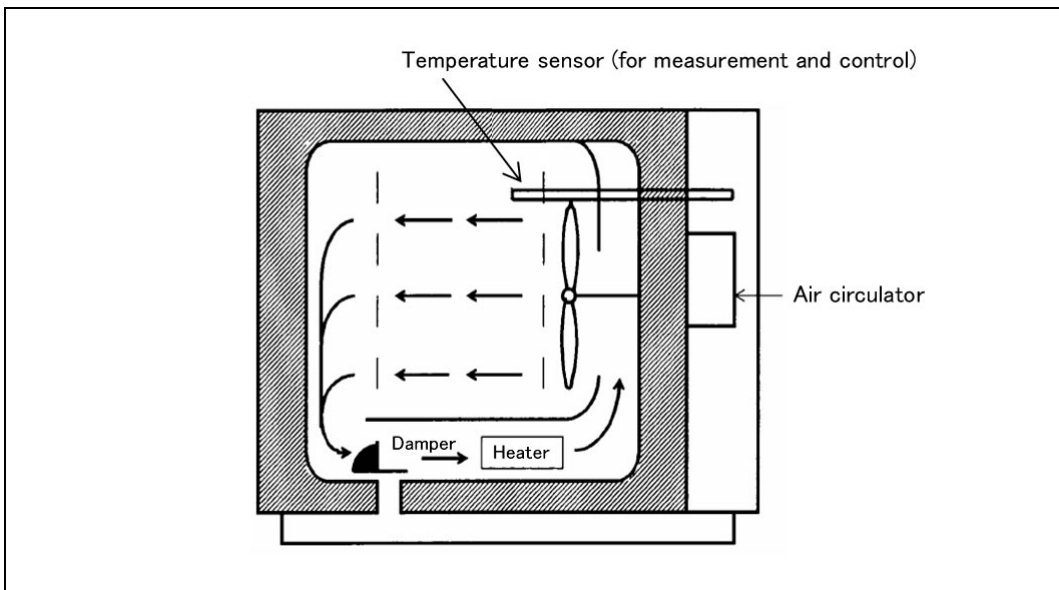


Fig.2 Example of temperature test chamber construction

As a result, in general some level of temperature deviation exists between the temperature at the center of the chamber and the setting temperature or indicated temperature on the test chamber temperature controllers. This occurs because of the chamber construction and the placement of the temperature sensors for measurement and control and the method of adjusting the temperature. Whether the chamber temperature achieved is at the temperature setting on the temperature controller is a control problem, and whether the temperature controller shows the true temperature at the center of the chamber is a measurement problem. In either case, if any temperature deviation exists between the temperature at the center of the chamber and the temperature controller, this temperature deviation should be seen as related to uncertainty. Compensation for the temperature deviation can be made by providing offset to the temperature controller, but uncertainty from the offset remains.

3-3 Method of expressing uncertainty

K 08 has a Budget Table of Uncertainty (a table listing uncertainty factors and estimates) requiring clarification of the limit to which uncertainty factors are included.

Ordinarily, through calibration and test verification the results show the size of the overall uncertainty but do not show the uncertainty factors nor the values for each factor. This standard, by providing examples of specific calculation procedures and values, has been able to clarify the factors related to uncertainty and their calculation.

4 Uncertainty factors and their method of evaluation

4-1 Analysis of uncertainty

Fig.3 shows the uncertainty factors of the temperature test chamber. The upper half charts the factors of uncertainty affecting the basic performance of the test chamber, and the lower half shows the uncertainty of the systems measuring that basic performance. When combining uncertainty, the independence of each factor is presumed. However, it is quite difficult to strictly prove whether the factors are independent.

The uncertainty inherent in temperature measurement systems used to measure test chamber temperature performance affects the measurements that are obtained. When using temperature measurement equipment with high precision responsiveness, the temperature fluctuation values obtained can be assumed to accurately portray the true conditions. On the other hand, when responsiveness and resolution are poor and temperature fluctuation cannot be detected, the fluctuation may increase greatly and include all sorts of margin of error. As a result, we must include evaluation of uncertainty based on temperature measurement equipment.

Measurement conditions also affect uncertainty, and measurement to evaluate uncertainty is performed according to the standard conditions stipulated in K 07.

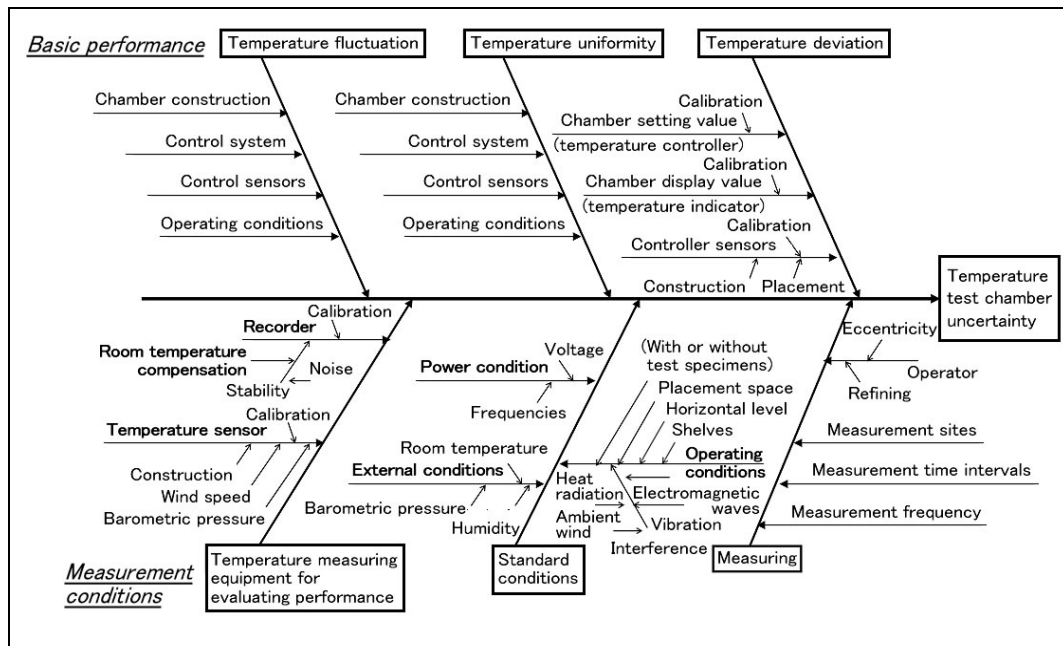


Fig.3 Uncertainty factors in the temperature test chamber

4-2 With or without test specimens

The manufacturer's guarantee of the temperature chamber performance refers, in principle, to a chamber with no specimens in the chamber. K 07 also stipulates performance evaluation methods with an empty chamber. Therefore, with K 08 we shall consider uncertainty with an empty chamber, and not take up conditions with specimens in the chamber. When specimens are in the chamber, uncertainty should be considered for each temperature (environment) test method.

4-3 Temperature fluctuation and temperature uniformity

When considering the temperature test chamber as a constant temperature chamber that has achieved an arbitrary uniform temperature, the uncertainty at that achieved temperature can be separated into the two factors of chronological temperature fluctuation in time, and spatial temperature fluctuation, referred to as temperature uniformity. IEC 60068-3-5 "Environmental testing – Part 3-5: Supporting documentation and guidance – Confirmation of the performance of temperature chambers" (below 60068-3-5) presents an extremely simplified concept diagram called "Examples of temperature differences." Fig.4 shows an excerpt of that diagram.

Temperature uniformity is represented through two methods: temperature variation in space based on the center of the working space (chamber center), and temperature gradient considering the entire working space. Standard 60068-3-5 does not specify measurement procedures for this temperature performance. K 07 specifically provides such evaluation and display methods.

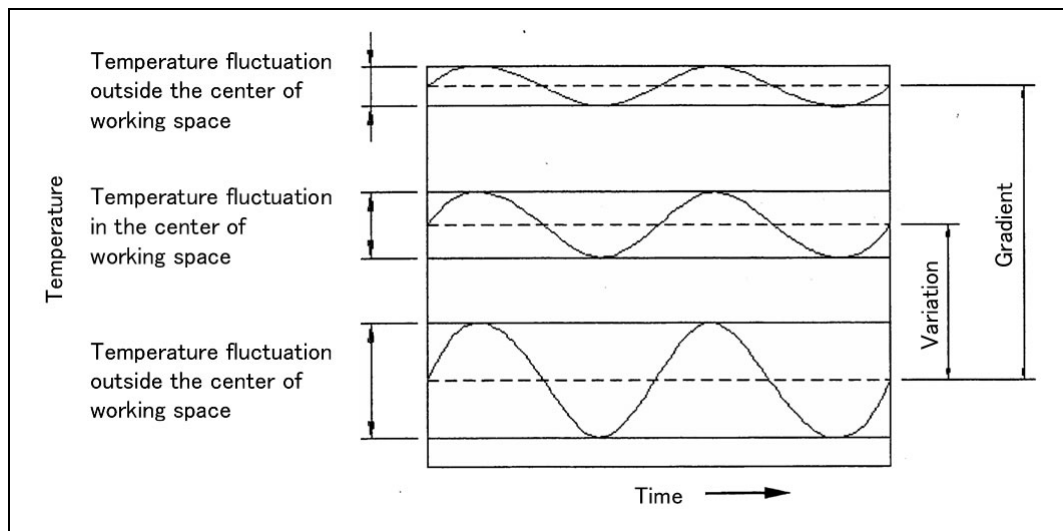


Fig.4 Example of temperature differences (from IEC 60068-3-5)

4-3-1 Uncertainty of temperature fluctuation (u_f)

For temperature fluctuation, K 07 stipulates finding the standard deviation σ_f for each of the measurement values obtained from temperature sensors placed at the measurement sites (generally nine, but this number can vary depending on the size of the chamber) in the working space, then multiplying by two and displaying as a $\pm$ value. The temperature fluctuation thus becomes $\pm 2\sigma_f$. The maximum value from among those obtained from the measurement sites is used to display temperature fluctuation. As a result, when using the standard uncertainty u_f ($^{\circ}\text{C}$) for temperature fluctuation found with K 08, σ_f is used without multiplying by two, so that $u_f = \sigma_f$.

4-3-2 Uncertainty of temperature uniformity (temperature variation in space) (u_v)

Chronological temperature fluctuation can be expected to be continually affecting the spatial temperature uniformity, but by averaging chronological fluctuations it is possible to regard temperature fluctuation and temperature uniformity as independent factors. In other words, by using the average temperatures measured at each measurement site at specified measurement intervals, we find the spatial temperature uniformity.

In K 07, temperature uniformity performance is specified as the two items of temperature gradient and temperature variation in space. K 07 defines temperature gradient as "the maximum difference between the average temperatures at two separate measurement points after temperature stability has been reached, at an arbitrary point in time in the working space." It defines temperature variation in space as "the difference between the average temperature in the center of the working space and at a separate arbitrary point in the working space after temperature stability has been reached."

Since the chamber center (the center of the working space) is considered to be the reference point in K 08, the uncertainty of temperature uniformity is found based on temperature variation in space. Since we can postulate uniform distribution in temperature variation from the chamber center to each apex of the working space, the temperature variation in space becomes δ_v ($^{\circ}\text{C}$), and with the standard uncertainty as u_v ($^{\circ}\text{C}$), we get $u_v = \delta_v / \sqrt{3}$.

4-4 Uncertainty from temperature variation from temperature setting (u_d)

The temperature variation between the center of the chamber and the setting on the temperature controller is not specified in K 07, and is not taken up in previous JTM standards. In other words, there is no clear specification value for the level of aberration of the temperature in the chamber from the temperature setting.

As a result, the temperature at the center of the chamber is actually measured with a separately prepared temperature indicator, and it is necessary to find the difference between the temperature variation of the setting temperature and the measured temperature inside the chamber. This temperature variation will vary depending on such factors as chamber construction, temperature control system, and external interference, but it can be considered to be a roughly systematic effect. With a uniform test temperature, one will get a roughly uniform value, and we can expect a comparatively greater value with a greater difference between the ambient temperature and the temperature setting. The greatest value will be seen at the highest temperature specification, and with the ambient temperature identical to the temperature setting we should see the lowest value. In other words, the supposition should hold true that there is a uniform distribution dependent upon the difference between the temperature setting and the ambient temperature. Therefore, room for debate exists concerning the method employed when changes occur in standard uncertainty.

As Fig.5 shows, the temperature variation from the temperature setting overlaps the temperature variation in space "a", and it would be possible to double count the uncertainty. A double count would not be expected to occur when temperature variation in space "a" is smaller than temperature variation in space "b", as the temperature uniformity (temperature gradient) is shown for the inside of the chamber that includes the temperature sensor.

In discussions on K 08, it was realized that the topic of discussion centered on whether this temperature variation included uncertainty. Ultimately, it was decided to rely on the decision of the person carrying out the uncertainty evaluation. However, since it was decided to show uncertainty also in the Budget Table, the limits of what is included in uncertainty is clarified.

The method of determining uncertainty is as follows: with the temperature variation set as the absolute value δ_d (°C), the standard uncertainty as u_d (°C), and assuming a uniform distribution, then $u_d = \delta_d / \sqrt{3}$.

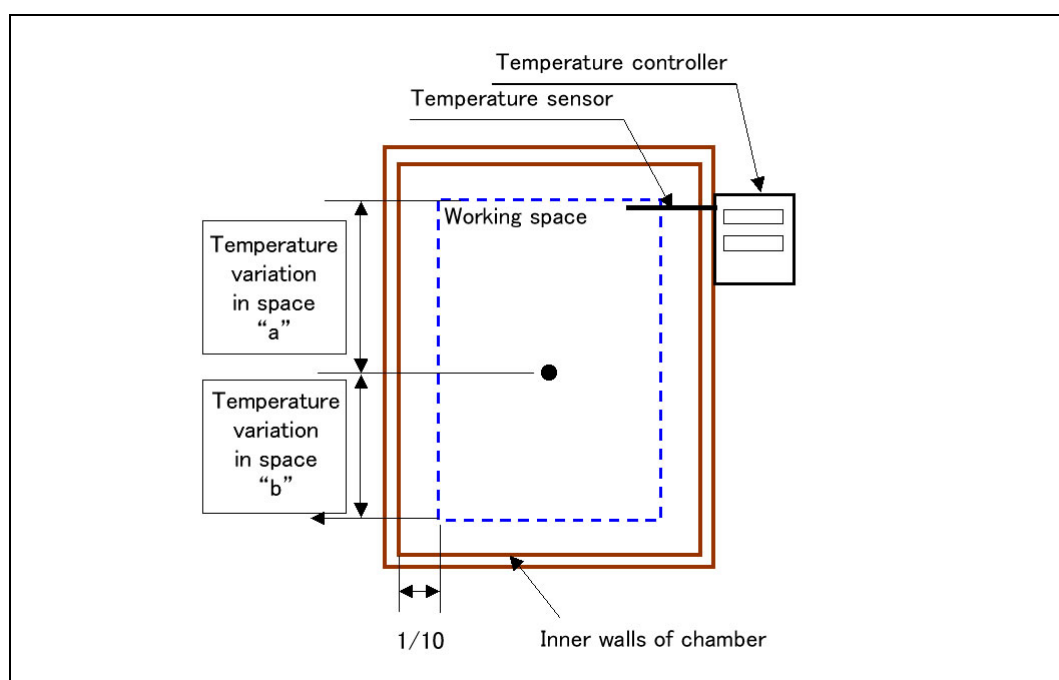


Fig.5 Temperature variation and temperature variation in space from temperature setting

4-5 Uncertainty from temperature indicator (u_i)

The thermocouples and resistance thermometers used as temperature sensors for evaluating temperature fluctuation and uniformity are generally connected to recording meters such as output converters and hybrid recorders, or possibly voltage and resistance detectors such as digital multimeters. As a result, the uncertainty of temperature indicators is combined with the uncertainty of these types of sensor and indicators.

It is assumed that these temperature indicators are calibrated. The calibration reports should contain uncertainty values, but if not, uncertainty can be estimated from the specification values and standard values. In addition, it would be desirable to evaluate reproducibility and stability, but when that is not possible, these factors can be estimated from the specification values. The range of the standard value and specification value $\pm "a"$ can be regarded as a uniform distribution and converted to standard uncertainty using $u_i = a/\sqrt{3}$.

5 Overall uncertainty

When estimating Type B^{*1} temperature uncertainty of the temperature test chamber from the catalog specification values according to K 07, the uncertainty as a temperature chamber is found by combining the registered temperature fluctuation and temperature variation in space. In addition, when performing a specific temperature test, throughout the test the temperature inside the chamber is measured with a specially provided thermometer, and the test temperature uncertainty can be found by combining the uncertainty of that thermometer.

To find Type A^{*2} temperature uncertainty of the temperature test chamber according to actual temperature measurements, add temperature fluctuation to temperature variation in space, and combine the uncertainty of the temperature sensor placed in the working space.

Whether the temperature variation between the center of the chamber and the temperature sensor should be handled as part of the uncertainty is left up to the judgment of the person who evaluates the uncertainty. However, since the uncertainty value is not unrelated to the whole, by attaching a Budget Table, one can indicate what factors have been included and precisely what uncertainty has been contributed. In addition, one must clarify whether that uncertainty is related to one temperature point or if it includes a temperature range.

By combining all of the above uncertainty factors, overall standard uncertainty "u" can be found with the following equation: $u = \sqrt{u_r^2 + u_v^2 + u_d^2 + u_i^2}$

Expanded uncertainty "U", with inclusive constant $k = 2$ becomes $U = 2 \times u$.

6 Example of calculating uncertainty

Table 2 shows measurement data for calculating uncertainty. This data consists of simulated values to aid in understanding, and differs from actual measured data. The Budget Tables for uncertainty from this data are shown in Tables 3 and 4. Table 3 is created without including the temperature variation from the temperature setting, while Table 4 includes that data.

Table 2 Temperature measurement data from inside the temperature test chamber

Temperature setting: 100°C										Units: °C
No.	Measurement sites	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Time (min.)					(Chamber center)				
0	0	99.9	99.7	99.8	100.3	100.4	100.3	100.1	100.5	100.7
1	2	99.8	99.6	99.8	100.3	100.4	100.2	100.1	100.4	100.8
2	4	99.9	99.5	99.8	100.3	100.4	100.2	100.1	100.4	100.5
3	6	99.8	99.4	99.8	100.3	100.4	100.2	100.1	100.4	100.6
4	8	99.9	99.7	99.8	100.3	100.4	100.2	100.1	100.4	100.6
5	10	99.9	99.6	99.8	100.3	100.3	100.2	100.1	100.4	100.7
6	12	99.9	99.5	99.8	100.3	100.4	100.2	100.1	100.4	100.8
7	14	99.8	99.4	99.8	100.3	100.4	100.2	100.1	100.4	100.7
8	16	99.9	99.7	99.8	100.2	100.3	100.2	100.1	100.4	100.6
9	18	99.9	99.6	99.8	100.2	100.2	100.2	100.1	100.4	100.4
10	20	99.9	99.5	99.8	100.3	100.1	100.2	100.1	100.4	100.3
11	22	99.9	99.4	99.8	100.3	100.0	100.2	100.1	100.4	100.4
12	24	99.8	99.7	99.8	100.3	100.1	100.2	100.1	100.4	100.4
13	26	99.8	99.6	99.8	100.2	100.1	100.2	100.1	100.4	100.5
14	28	99.8	99.5	99.8	100.2	100.2	100.1	100.0	100.4	100.5
15	30	99.8	99.4	99.8	100.2	100.2	100.2	100.0	100.4	100.6
A	Average temperature	99.86	99.55	99.80	100.27	100.27	100.20	100.09	100.41	100.57
B	Maximum temperature	99.9	99.7	99.8	100.3	100.4	100.3	100.1	100.5	100.8
C	Minimum temperature	99.8	99.4	99.8	100.2	100.0	100.1	100.0	100.4	100.3
D	Temperature fluctuation (B - C)	0.1	0.3	0.0	0.1	0.4	0.2	0.1	0.1	0.5
E	Difference from average temp. at chamber center (temperature variation in space)	-0.41	-0.72	-0.47	0.00	-	-0.07	-0.18	0.14	0.30

Table 3 Budget Table for uncertainty Temperature setting: 100°C
(Temperature variation from temperature setting not included)

Units: °C

Uncertainty factors and symbols	Symbol and calculation	Size of factor	Standard uncertainty	Combined standard uncertainty	Comments
Temperature fluctuation	δ_f	0.3			
(i) Uncertainty from temperature fluctuation: u_f	$\delta_f / 2$		0.15		
Temperature variation in space	δ_v	0.72			
(ii) Uncertainty from temperature variation in space: u_v	$\delta_v / \sqrt{3}$		0.416		
Chamber uncertainty	$\sqrt{(u_f^2 + u_v^2)}$			0.442	Combination of (i) and (ii)
Temperature sensor (Thermocouple type T [Copper/Copper-Nickel], class 1)	t_r	0.5			Permissible deviation of IEC 584 thermocouples
(iii) Uncertainty from temperature sensor: u_s	$t_r / \sqrt{3}$		0.289		
Temperature converter (recorder)	R_a	0.55			Specification precision
(iv) Uncertainty from temperature converter: u_{r1}	$R_a / \sqrt{3}$		0.318		
Thermocouple compensator (room temperature compensator)	R_j	0.5			Specification precision
(v) Uncertainty from thermocouple compensator: u_{r2}	$R_j / \sqrt{3}$		0.289		
Uncertainty from temperature indicator: u_i	$\sqrt{(u_s^2 + u_{r1}^2 + u_{r2}^2)}$			0.517	Combination of (iii) to (v)
Standard uncertainty for chamber performance: u	$\sqrt{(u_f^2 + u_v^2 + u_s^2 + u_{r1}^2 + u_{r2}^2)}$			0.68	Total combined (i) to (v)
Expanded uncertainty for chamber performance: U ($k=2$)	$2 * u$			1.4	

Table 4 Budget Table for uncertainty Temperature setting: 100°C
(Temperature variation from temperature setting included)

Units: °C

Uncertainty factors and symbols	Symbol and calculation	Size of factor	Standard uncertainty	Combined standard uncertainty	Comments
Temperature fluctuation	δ_f	0.3			
(i) Uncertainty from temperature fluctuation: u_f	$\delta_f / 2$		0.150		
Temperature variation in space	δ_v	0.72			
(ii) Uncertainty from temperature variation in space: u_v	$\delta_v / \sqrt{3}$		0.416		
Temperature variation from temperature setting	δ_d	0.27			
(iii) Uncertainty from temperature variation from temperature setting: u_d	$\delta_d / \sqrt{3}$		0.156		
Chamber uncertainty	$\sqrt{(u_f^2 + u_v^2 + u_d^2)}$			0.469	Combination of (i) and (iii)
Temperature sensor (Thermocouple type T [Copper/Copper-Nickel], class 1)	t_r	0.5			Permissible deviation of IEC 584 thermocouples
(iv) Uncertainty from temperature sensor: u_s	$t_r / \sqrt{3}$		0.289		
Temperature converter (recorder)	R_a	0.55			Specification precision
(v) Uncertainty from temperature converter: u_{r1}	$R_a / \sqrt{3}$		0.318		
Thermocouple compensator (room temperature compensator)	R_j	0.5			Specification precision
(vi) Uncertainty from thermocouple compensator: u_{r2}	$R_j / \sqrt{3}$		0.289		
Uncertainty from temperature indicator: u_i	$\sqrt{(u_s^2 + u_{r1}^2 + u_{r2}^2)}$			0.517	Combination of (iv) to (vi)
Standard uncertainty for chamber performance: u	$\sqrt{(u_f^2 + u_v^2 + u_d^2 + u_s^2 + u_{r1}^2 + u_{r2}^2)}$			0.698	Total combined (i) to (vi)
Expanded uncertainty for chamber performance: U ($k=2$)	$2 * u$			1.4	

7 Conclusion

K 08 has set out to make it easy to apply, as concretely as possible, the method of finding test temperature chamber uncertainty. This provides not only calculation procedures, but simulated data and budget tables as well. Using this, anyone should easily be able to find uncertainty values.

As a standard, K 07 is used to find uncertainty in performance of the total working space guaranteed by the manufacturer. When the user actually operates the equipment, the true importance of performance is with specimens placed in the chamber. Uncertainty can be evaluated based on this standard for that case as well.

Currently, JTM is deliberating the method of evaluating performance for temperature and humidity chambers (rooms), and following that, JTM plans to create a standard for evaluating uncertainty of temperature and humidity chambers.

[Terminology]

*1. Type B uncertainty

The following are examples of Type B uncertainty: uncertainty based on a series of observed values not statistically analyzed; uncertainty concretely documented in calibration reports; uncertainty based on catalog specifications and standard values.

*2. Type A uncertainty

Uncertainty based on statistical analysis of a series of observed values.